Machine learning for predicting sediment particle size distributions

UE24CS352A Machine Learning · Mini-project 62 · Team B-16
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3,998	**9**	**0.86 / 0.82**	**0.60 / 0.62**
samples from 92 casts	input features	R^2 (d50 / σ), random split	R^2 (d50 / σ), unseen casts

1. Problem statement

Mud particles suspended in estuaries clump into flocs and break apart again within seconds. Their size sets how fast they settle, the most uncertain input to sediment transport models, and no general physical model predicts it well. Following Egan (Stanford CS229, 2019), we ask whether regression can predict two summaries of the particle size distribution from routine water measurements: the **median diameter d50** and the **spread σ** (the square root of the paper's variance). Success is measured by R^2 on held-out data.

2. Dataset

The paper's data are not public, so we used the closest public equivalent: a U.S. Geological Survey release from the same bay and instrument family (doi:10.5066/P99Q4CHM).

Coverage	Vertical casts at a deep channel site (11–14 m) and a shallows site (2–5 m) in south San Francisco Bay on 21, 22, 24 and 28 July 2020.
Instruments	LISST-200X laser particle sizer: volume in 36 size classes (1.2–459 µm), temperature, beam attenuation. RBR CTD: salinity, turbidity, depth.
Size	94 LISST files (45,639 one-second samples) and 99 CTD files. After cleaning: 3,998 ten-second bursts from 92 casts (50 channel, 42 shallows).
Against the paper	We have salinity and temperature, but not chlorophyll, refractive index or wave and tidal velocity.

3. Approach

- **Features (9).** Sensor depth, height above bed, water depth (tide stage), tide rate, temperature, salinity, turbidity, beam attenuation and site.
- **Models.** The paper's Random Forest and support vector regression (SVR), plus a ridge baseline, gradient boosting and a 10-unit neural network. As in the paper, forest size (50 trees) comes from the out-of-bag curve and SVR's C and ε from a validation sweep.
- **Evaluation, our main extension.** Consecutive samples in a cast are nearly identical, so a random split puts near-copies of each test point in training. We test four ways: the paper's random 70/30 split, whole casts held out, a whole day held out, and train on one site, test on the other.

4. Implementation

build_dataset.py	Reads the NetCDF casts, corrects the 3–7 s clock offset between the two instruments, removes bad samples and averages into 10-second bursts.
psd.py	Computes d50 and σ from each size spectrum.
train_evaluate.py	Feature importance, tuning, five models under four test protocols.
demo.py	Holds out one cast, retrains and compares predicted with measured values.

5. Results

R² on test data, averaged over splits. Higher is better; below 0 is worse than always predicting the mean. The shaded columns are the paper's protocol.

Model	Random 70/30 (paper)		Unseen casts		Unseen day		Unseen site	
	d50	σ	d50	σ	d50	σ	d50	σ
Ridge (linear baseline)	0.43	0.53	0.36	0.45	0.19	0.31	-5.39	< −10
Random Forest	0.86	0.82	0.56	0.58	0.35	0.46	-1.67	-0.95
SVR (RBF)	0.69	0.72	0.60	0.62	0.45	0.53	-1.90	-3.83
Gradient Boosting	0.85	0.81	0.59	0.60	0.42	0.50	-2.32	-1.17
Neural net (10 units)	0.69	0.72	0.53	0.62	0.34	0.52	< −10	-7.09

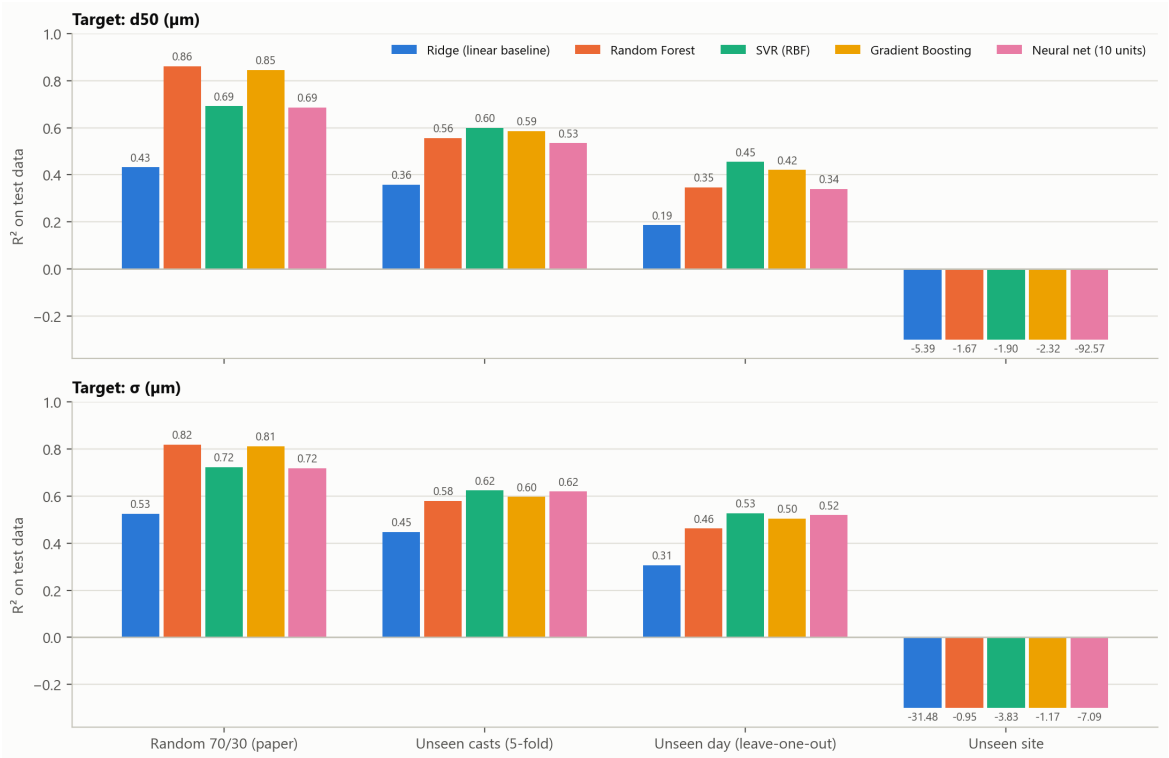


Figure 1. R² of every model under the four test protocols. Bars are cut off at −0.3; the labels give the true values.

- **The paper's result reproduces.** On a random split, Random Forest reaches R² = 0.86 (d50) and 0.82 (σ); the paper reports 0.81 and 0.75.
- **Part of that score is memorised neighbours.** On unseen casts Random Forest falls to 0.56 and 0.58, and the smoother SVR becomes the best model (0.60 and 0.62).
- **Nothing transfers between sites.** Every model scores below zero, because the two sites barely overlap in water depth or particle size.
- **Salinity is not the main driver here.** Depth, tide stage and beam attenuation matter most. Salinity carries 8 % of the importance against 72 % in the paper; it only ranged from 30.1 to 32.3 psu. Adding salinity and turbidity lifts unseen-cast R² from 0.54 / 0.58 to 0.56 / 0.58.

6. Conclusions

Machine learning explains about 56–62 % of the variation in floc size and spread on new casts: useful, but short of the 0.82–0.86 a random split suggests. For autocorrelated field data the split decides the score, so the paper's worry about overfitting was justified. Our limits are four days in one summer, two sites and a narrow salinity range. The next step is to add current and wave measurements and to train across more sites and seasons.